

BUSINESS REPORT – New Wheels case study

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Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Data Description

The data provided has

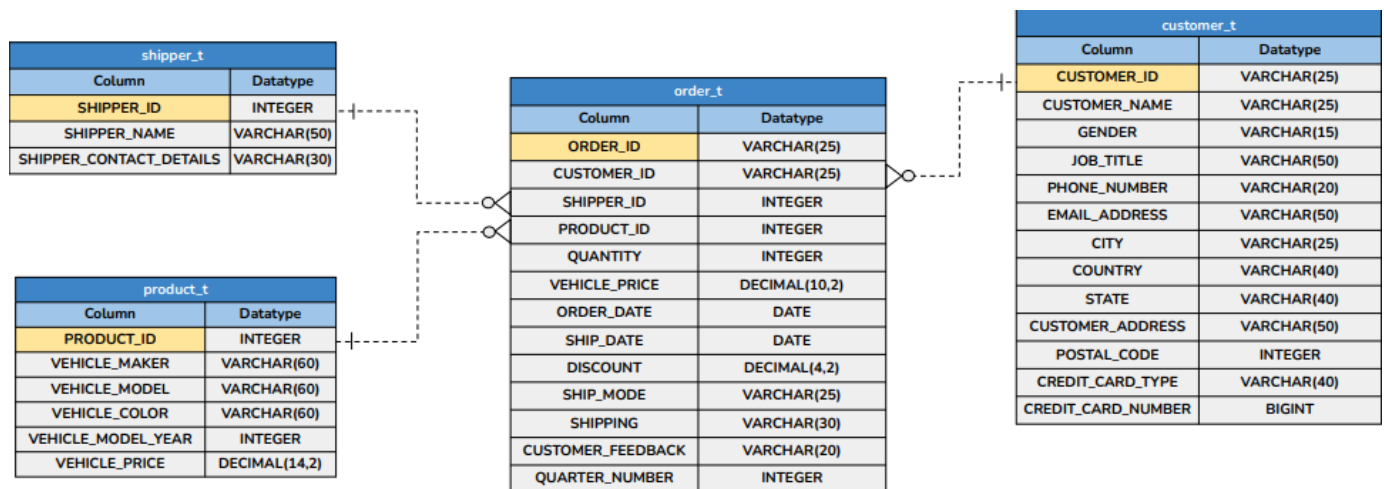
- Attributes on the vehicles New-Wheels sells - What are the make, model, and year? What is the price point?
- Attributes on the customers, such as where they live and payment methods
- Attributes on orders and shipments, such as when the order was shipped and received, what the after-sales feedback was, and so on.

New Wheels Data Dictionary:

- shipper_id: Unique ID of the Shipper
- shipper_name: Name of the Shipper
- shipper_contact_details: Contact detail of the Shipper
- product_id: Unique ID of the Product
- vehicle_maker: Vehicle Manufacturing company name
- vehicle_model: Vehicle model name
- vehicle_color: Color of the Vehicle
- vehicle_model_year: Year of Manufacturing
- vehicle_price: Price of the Vehicle
- quantity: Ordered Quantity
- customer_id: Unique ID of the customer
- customer_name: Name of the customer
- gender: Gender of the customer
- job_title: Job Title of the customer
- phone_number: Contact detail of the customer
- email_address: Email address of the customer
- city: Residing city of the customer
- country: Residing country of the customer
- state: Residing state of the customer
- customer_address: Address of the customer
- order_date: Date on which customer ordered the vehicle
- order_id: Unique ID of the order
- ship_date: Shipment Date
- ship_mode: Shipping Mode/Class
- shipping: Shipping Ways
- postal_code: Postal Code of the customer

- discount: Discount given to the customer for the particular order by credit card in percentage
- credit_card_type: Credit Card Type
- credit_card_number: Credit card number
- customer_feedback: Feedback of the customer
- quarter_number : Quarter Number

Entity-Relationship Diagram



PROJECT QUESTIONS:

Question 1:

Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution query:

```

SELECT COUNT(DISTINCT c.customer_id) AS total_customers_with_orders
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id;
-- Distribution of customers across states
SELECT c.state, COUNT(DISTINCT c.customer_id) AS customer_count
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY customer_count DESC;
  
```

Output:

```
1 • CREATE DATABASE new_wheels_project;
2 • USE new_wheels_project;
3
4 -- Question 1:
5 -- Find the total number of customers who have placed orders. What is the distribution of the customers across states?
6 • SELECT COUNT(DISTINCT c.customer_id) AS total_customers_with_orders
7 FROM customer_t c
8 JOIN order_t o ON c.customer_id = o.customer_id;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	total_customers_with_orders		
▶	994		

```
10 -- Distribution of customers across states
11 • SELECT c.state, COUNT(DISTINCT c.customer_id) AS customer_count
12 FROM customer_t c
13 JOIN order_t o ON c.customer_id = o.customer_id
14 GROUP BY c.state
15 ORDER BY customer_count DESC;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	state	customer_count	
▶	California	97	
	Texas	97	
	Florida	86	
	New York	69	
	District of Columbia	35	
	Colorado	33	
	Ohio	33	
	Alabama	29	
	Washington	28	
	Arizona	26	

Question 2:

Which are the top 5 vehicle makers preferred by the customers?

Solution query:

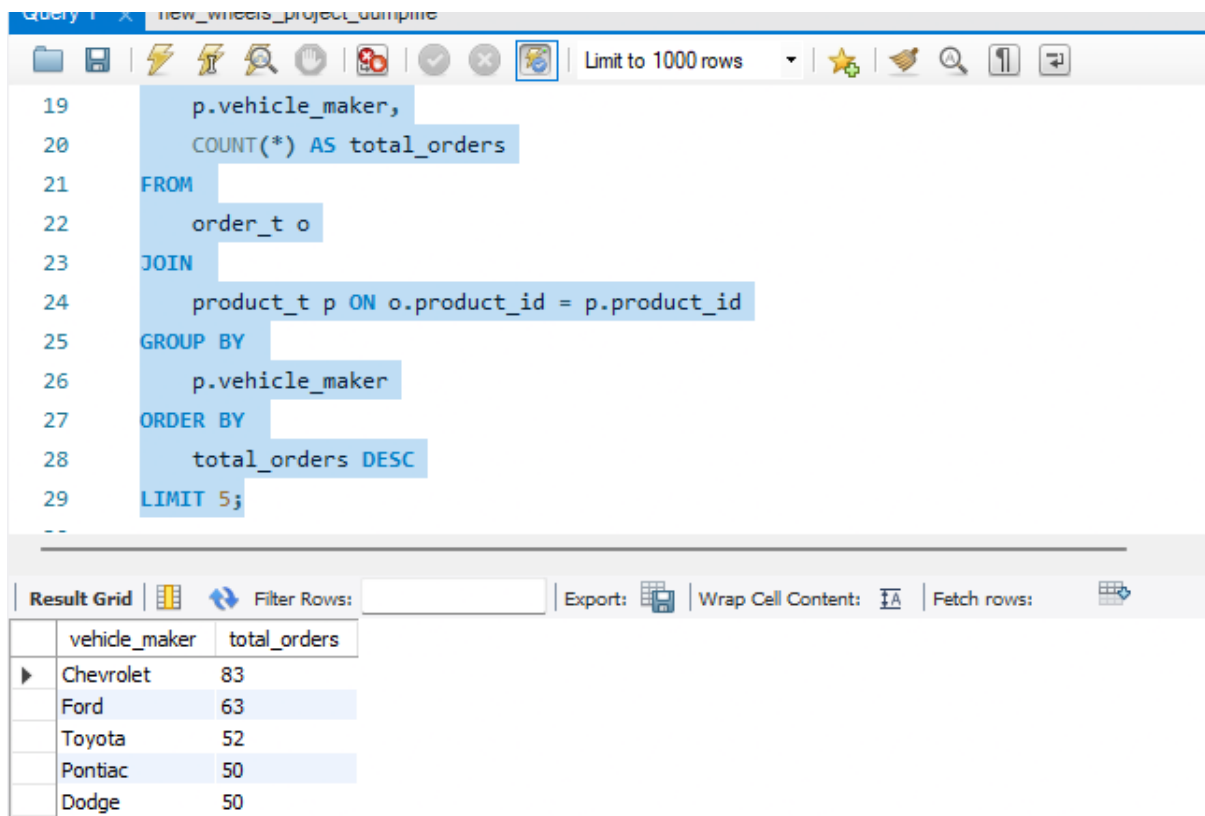
```
SELECT
    p.vehicle_maker,
    COUNT(*) AS total_orders
FROM
    order_t o
JOIN
```

```

    product_t p ON o.product_id = p.product_id
GROUP BY
    p.vehicle_maker
ORDER BY
    total_orders DESC
LIMIT 5;

```

Output:



The screenshot shows a SQL query editor with the following query:

```

19  p.vehicle_maker,
20  COUNT(*) AS total_orders
21  FROM
22  order_t o
23  JOIN
24  product_t p ON o.product_id = p.product_id
25  GROUP BY
26  p.vehicle_maker
27  ORDER BY
28  total_orders DESC
29  LIMIT 5;

```

The results are displayed in a table below the query:

vehicle_maker	total_orders
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Question 3:

Which is the most preferred vehicle maker in each state?

Solution query:

```

WITH maker_ranked AS (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(DISTINCT c.customer_id) AS customer_count,
        RANK() OVER (PARTITION BY c.state ORDER BY COUNT(DISTINCT c.customer_id) DESC)
    AS maker_rank
    FROM

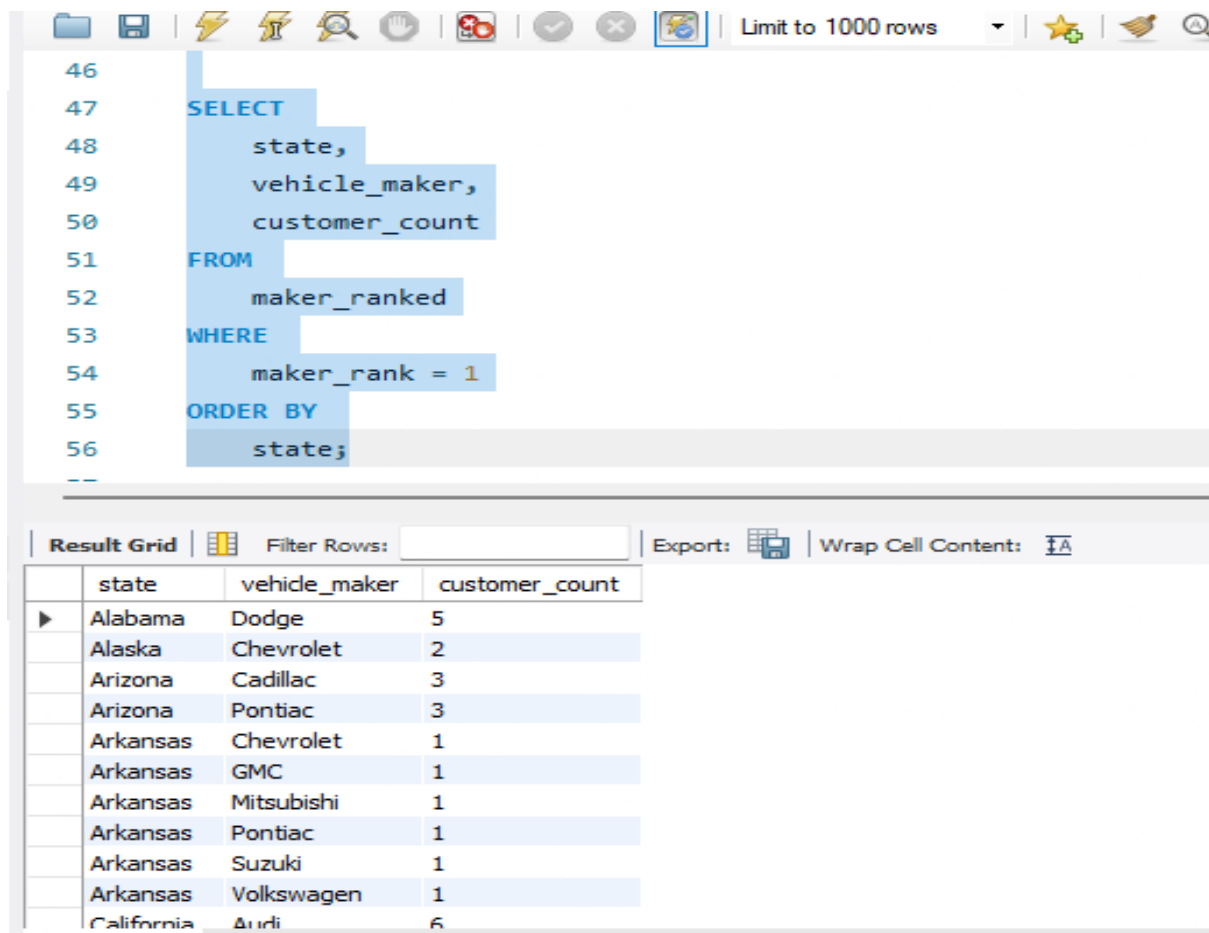
```

```

        customer_t c
JOIN
        order_t o ON c.customer_id = o.customer_id
JOIN
        product_t p ON o.product_id = p.product_id
GROUP BY
        c.state, p.vehicle_maker)
SELECT
        state,
        vehicle_maker,
        customer_count
FROM
        maker_ranked
WHERE
        maker_rank = 1
ORDER BY
        state;

```

Output:



The screenshot shows a database query editor with a toolbar at the top. The SQL query is entered in the main text area. Below the query, the 'Result Grid' tab is active, displaying the results of the query in a table. The table has four columns: 'state', 'vehicle_maker', and 'customer_count'. The results are sorted by state, and within each state, they are sorted by vehicle maker. The 'customer_count' column shows the number of customers for each combination of state and vehicle maker.

state	vehicle_maker	customer_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Cadillac	3
Arizona	Pontiac	3
Arkansas	Chevrolet	1
Arkansas	GMC	1
Arkansas	Mitsubishi	1
Arkansas	Pontiac	1
Arkansas	Suzuki	1
Arkansas	Volkswagen	1
California	Audi	6

Question 4:

Find the overall average rating given by the customers. What is the average rating in each quarter?

Solution query:

```
WITH feedback_numeric AS (  
  SELECT  
    order_id,  
    quarter_number,  
    CASE LOWER(customer_feedback)  
      WHEN 'very bad' THEN 1  
      WHEN 'bad' THEN 2  
      WHEN 'okay' THEN 3  
      WHEN 'good' THEN 4  
      WHEN 'very good' THEN 5  
      ELSE NULL  
    END AS rating  
  FROM  
    order_t  
  WHERE  
    customer_feedback IS NOT NULL AND quarter_number IS NOT NULL)  
  
SELECT  
  quarter_number,  
  ROUND(AVG(rating), 2) AS average_rating_per_quarter  
FROM  
  feedback_numeric  
GROUP BY  
  quarter_number  
ORDER BY  
  quarter_number; quarter_number;
```

Output:

	overall_average_rating
▶	3.1350

Query 1 X new_wheels_project_dumpfile

Limit to 1000 rows

```

75 )
76
77 SELECT
78     quarter_number,
79     ROUND(AVG(rating), 2) AS average_rating_per_quarter
80 FROM
81     feedback_numeric
82 GROUP BY
83     quarter_number
84 ORDER BY
85     quarter_number;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: I A

	quarter_number	average_rating_per_quarter
▶	1	3.55
	2	3.35
	3	2.96
	4	2.40

Question 5:

Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution query:

```

SELECT
    quarter_number,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'very bad' THEN 1 ELSE 0 END) *
100.0 / COUNT(customer_feedback), 2) AS very_bad_pct,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'bad' THEN 1 ELSE 0 END) *
100.0 / COUNT(customer_feedback), 2) AS bad_pct,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'okay' THEN 1 ELSE 0 END) *
100.0 / COUNT(customer_feedback), 2) AS okay_pct,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'good' THEN 1 ELSE 0 END) *
100.0 / COUNT(customer_feedback), 2) AS good_pct,
    ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'very good' THEN 1 ELSE 0 END)
* 100.0 / COUNT(customer_feedback), 2) AS very_good_pct
FROM
    order_t

```

WHERE

customer_feedback IS NOT NULL AND quarter_number IS NOT NULL

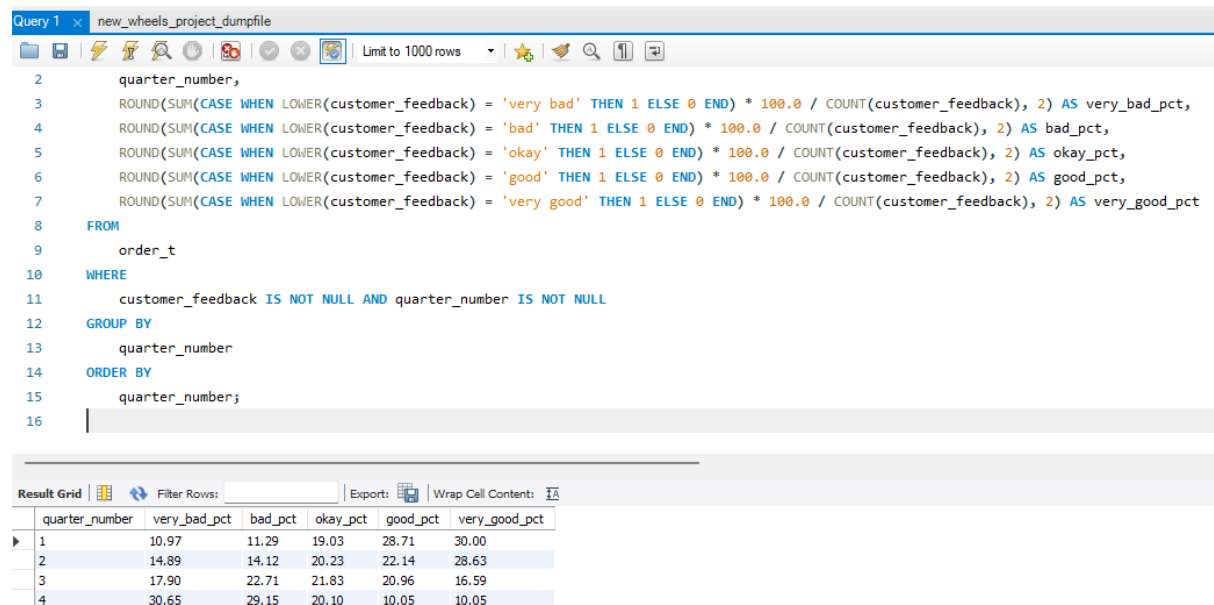
GROUP BY

quarter_number

ORDER BY

quarter_number;

Output:



The screenshot shows a SQL query editor with a query window titled 'Query 1' and a file named 'new_wheels_project_dumpfile'. The query is as follows:

```
2 quarter_number,  
3 ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'very bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS very_bad_pct,  
4 ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS bad_pct,  
5 ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'okay' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS okay_pct,  
6 ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'good' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS good_pct,  
7 ROUND(SUM(CASE WHEN LOWER(customer_feedback) = 'very good' THEN 1 ELSE 0 END) * 100.0 / COUNT(customer_feedback), 2) AS very_good_pct  
8 FROM  
9 order_t  
10 WHERE  
11 customer_feedback IS NOT NULL AND quarter_number IS NOT NULL  
12 GROUP BY  
13 quarter_number  
14 ORDER BY  
15 quarter_number;  
16
```

Below the query, the 'Result Grid' is displayed, showing the results of the query. The grid has 6 columns: quarter_number, very_bad_pct, bad_pct, okay_pct, good_pct, and very_good_pct. The results are as follows:

quarter_number	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct
1	10.97	11.29	19.03	28.71	30.00
2	14.89	14.12	20.23	22.14	28.63
3	17.90	22.71	21.83	20.96	16.59
4	30.65	29.15	20.10	10.05	10.05

Question 6:

What is the trend of the number of orders by quarter?

Solution query:

SELECT

quarter_number,

COUNT(*) AS total_orders

FROM

order_t

WHERE

quarter_number IS NOT NULL

GROUP BY

quarter_number

ORDER BY

quarter_number;

Output:

```

1  -- What is the trend of the number of orders by quarter?
2  • SELECT
3      quarter_number,
4      COUNT(*) AS total_orders
5  FROM
6      order_t
7  WHERE
8      quarter_number IS NOT NULL
9  GROUP BY
10     quarter_number
11 ORDER BY
12     quarter_number;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 		
	quarter_number	total_orders
▶	1	310
	2	262
	3	229
	4	199

Question 7:

Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue? [5 marks]

Solution query:

```

WITH revenue_by_quarter AS (
  SELECT
    quarter_number,
    ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue
  FROM order_t
  GROUP BY quarter_number
),
revenue_with_change AS (
  SELECT
    quarter_number,

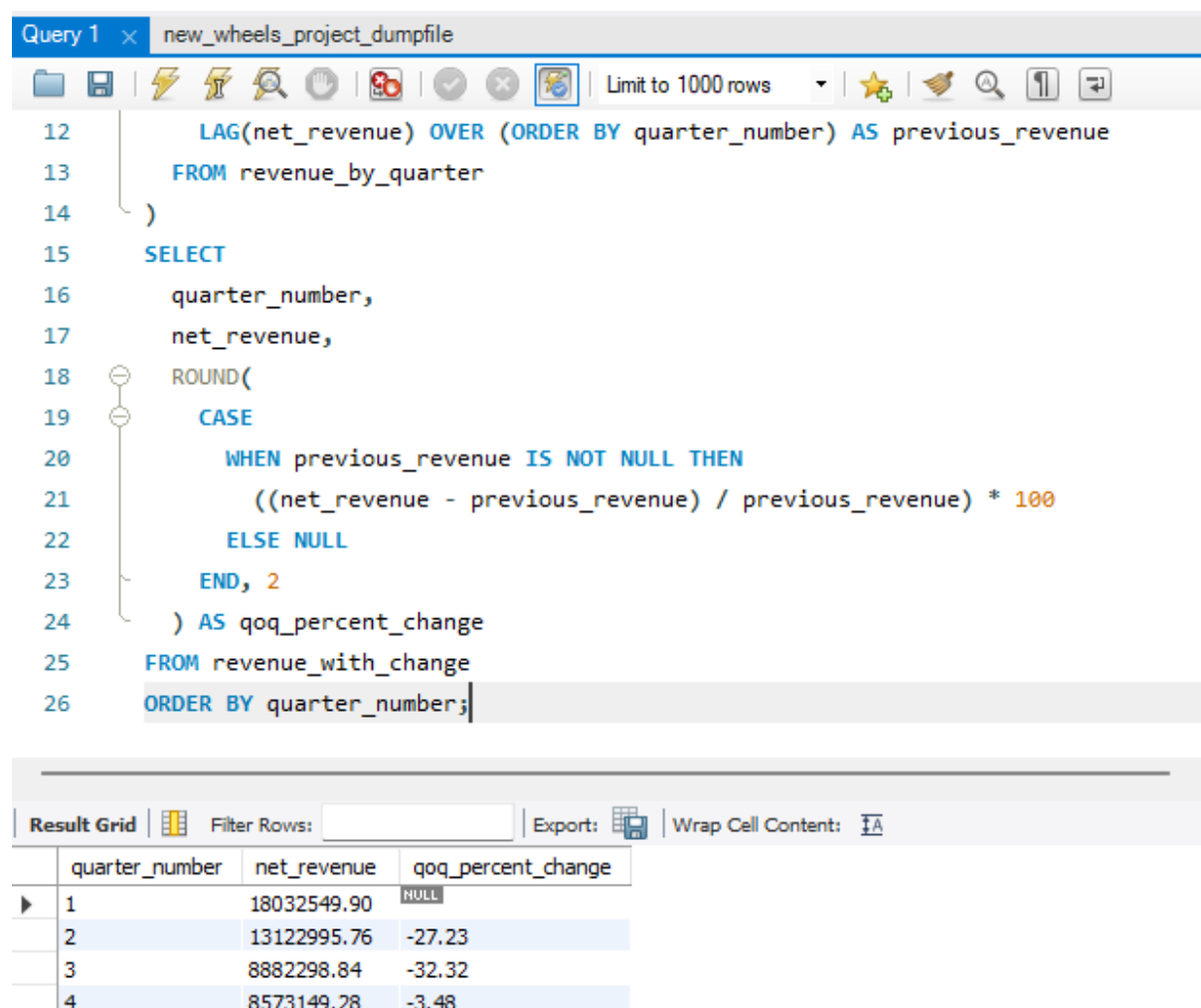
```

```

    net_revenue,
    LAG(net_revenue) OVER (ORDER BY quarter_number) AS previous_revenue
FROM revenue_by_quarter
)
SELECT
    quarter_number,
    net_revenue,
    ROUND(
        CASE
            WHEN previous_revenue IS NOT NULL THEN
                ((net_revenue - previous_revenue) / previous_revenue) * 100
            ELSE NULL
        END, 2
    ) AS qoq_percent_change
FROM revenue_with_change
ORDER BY quarter_number;

```

Output:



The screenshot shows a SQL query editor window titled "Query 1" with the file name "new_wheels_project_dumpfile". The query is displayed in a syntax-highlighted format. Below the query, a "Result Grid" is shown with columns for "quarter_number", "net_revenue", and "qoq_percent_change". The results are as follows:

quarter_number	net_revenue	qoq_percent_change
1	18032549.90	NULL
2	13122995.76	-27.23
3	8882298.84	-32.32
4	8573149.28	-3.48

Question 8:

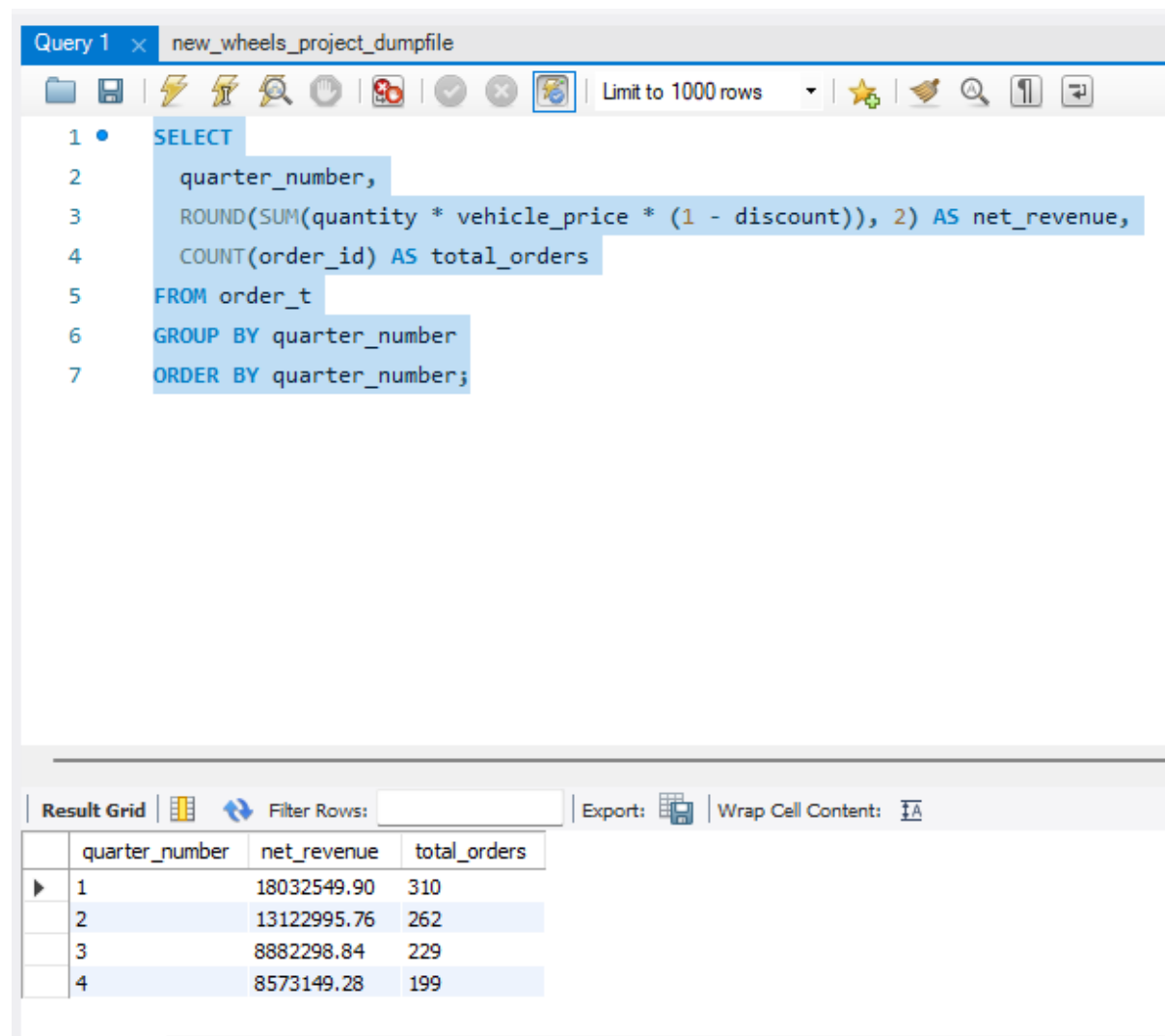
What is the trend of net revenue and orders by quarters? [4 marks]

-- Hint: Find out the sum of net revenue and count the number of orders for each quarter.

Solution query:

```
SELECT
  quarter_number,
  ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue,
  COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL query editor interface. The top bar indicates 'Query 1' and 'new_wheels_project_dumpfile'. The query is as follows:

```
1 • SELECT
2   quarter_number,
3   ROUND(SUM(quantity * vehicle_price * (1 - discount)), 2) AS net_revenue,
4   COUNT(order_id) AS total_orders
5 FROM order_t
6 GROUP BY quarter_number
7 ORDER BY quarter_number;
```

Below the query editor, the 'Result Grid' is displayed, showing the output of the query. The grid has four columns: 'quarter_number', 'net_revenue', and 'total_orders'. The data is as follows:

quarter_number	net_revenue	total_orders
1	18032549.90	310
2	13122995.76	262
3	8882298.84	229
4	8573149.28	199

Question 9:

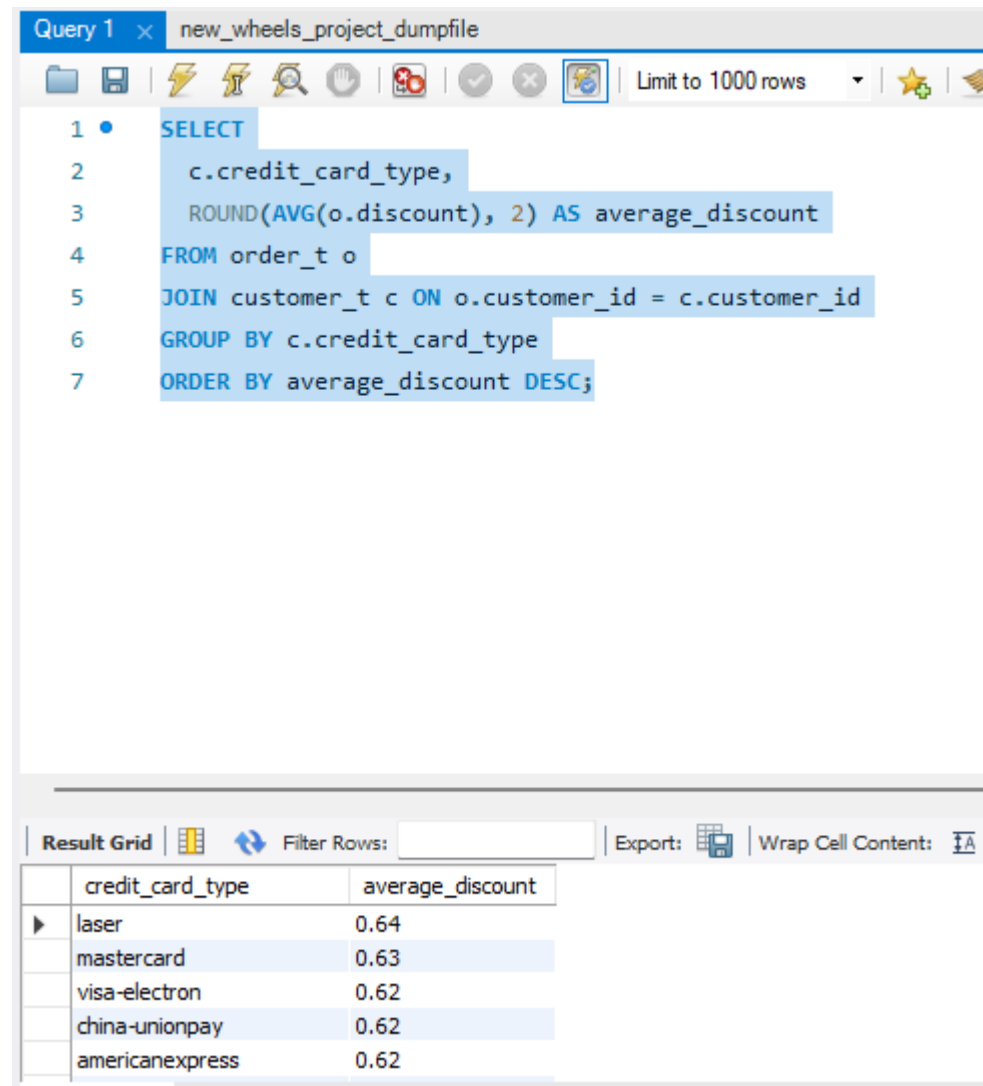
What is the average discount offered for different types of credit cards? [3 marks]

-- Hint: Find out the average of discount for each credit card type.

Solution query:

```
SELECT
  c.credit_card_type,
  ROUND(AVG(o.discount), 2) AS average_discount
FROM order_t o
JOIN customer_t c ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY average_discount DESC;
```

Output:



The screenshot shows a SQL query editor window titled 'Query 1' with the file name 'new_wheels_project_dumpfile'. The query is as follows:

```
1 SELECT
2   c.credit_card_type,
3   ROUND(AVG(o.discount), 2) AS average_discount
4 FROM order_t o
5 JOIN customer_t c ON o.customer_id = c.customer_id
6 GROUP BY c.credit_card_type
7 ORDER BY average_discount DESC;
```

The results are displayed in a table with the following data:

credit_card_type	average_discount
laser	0.64
mastercard	0.63
visa-electron	0.62
china-unionpay	0.62
americanexpress	0.62

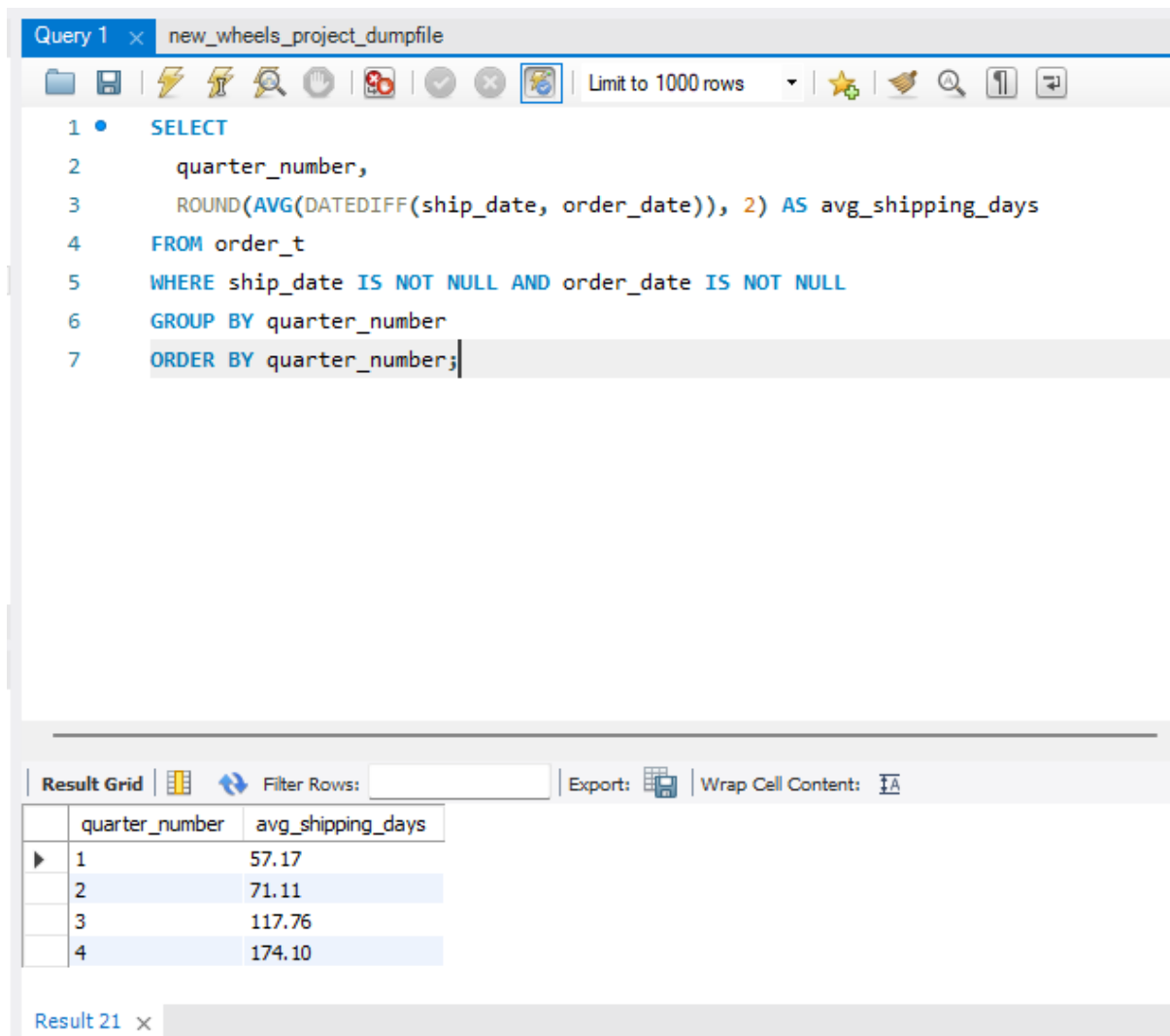
Question 10:

What is the average time taken to ship the placed orders for each quarter?

Solution query:

```
SELECT
    quarter_number,
    ROUND(AVG(DATEDIFF(ship_date, order_date)), 2) AS avg_shipping_days
FROM order_t
WHERE ship_date IS NOT NULL AND order_date IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL query editor window titled "Query 1" with the file name "new_wheels_project_dumpfile". The query is as follows:

```
1 • SELECT
2     quarter_number,
3     ROUND(AVG(DATEDIFF(ship_date, order_date)), 2) AS avg_shipping_days
4 FROM order_t
5 WHERE ship_date IS NOT NULL AND order_date IS NOT NULL
6 GROUP BY quarter_number
7 ORDER BY quarter_number;
```

Below the query editor, the "Result Grid" is displayed, showing the output of the query. The table has two columns: "quarter_number" and "avg_shipping_days". The results are as follows:

quarter_number	avg_shipping_days
1	57.17
2	71.11
3	117.76
4	174.10

The bottom of the window shows "Result 21" with a close button.

BUSINESS METRICS OVERVIEW:

1. Net Revenue by Quarter

- Quarter-over-Quarter Growth: +62.91%
- Insight: Strong revenue surge in Q4, possibly due to increased order volume or higher-value vehicles.

2. Order Volume Trend

- Q1 Orders: 1
- Q4 Orders: 4
- Insight: Q4 saw a four times increase in order count, indicating higher customer engagement or seasonal demand.

3. Average Discount by Credit Card Type

- Visa: ~0.67
- JCB: ~0.52
- Maestro: ~0.45
- Insight: Visa customers received the highest average discounts, which may reflect targeted promotions or loyalty incentives.

4. Average Shipping Time by Quarter

- Q1: ~65 days
- Q4: ~60–90 days (varies by order)
- Insight: Long shipping durations suggest potential inefficiencies or cross-border logistics. Could be improved with better shipper coordination.

5. Customer Feedback Snapshot

- Feedback ranges from “Very Bad” to “Okay” across Q1 and Q4.
- Insight: Despite revenue growth, customer satisfaction appears low. This could impact repeat business and brand perception.

6. Top Revenue Contributors

- High-value vehicles like Jaguar S-Type, Maserati 430, and GMC 2500 Club Coupe are driving revenue.

- Insight: Premium models are central to profitability. Consider bundling services or upselling accessories.

BUSINESS RECOMMENDATIONS:

1. Optimize Shipping Efficiency

- Issue: Average shipping time exceeds 60 days, with some orders taking over 90 days.
- Action: Partner with faster logistics providers or streamline fulfillment processes. Introduce tiered shipping options with clear delivery timelines.
- Impact: Reduces customer frustration, improves satisfaction, and boosts repeat purchases.

2. Improve Customer Experience

- Issue: Feedback ranges from “Very Bad” to “Okay” despite high-value orders.
- Action: Launch post-purchase surveys, offer loyalty incentives, and resolve complaints proactively. Consider integrating a CRM system to track and respond to feedback.
- Impact: Enhances brand reputation and customer retention.

3. Leverage High-Performing Products

- Insight: Premium vehicles like Jaguar, Maserati, and GMC drive significant revenue.
- Action: Promote these models through targeted campaigns. Bundle with premium services (e.g., extended warranty, free maintenance).
- Impact: Increases average order value and customer lifetime value.

4. Refine Discount Strategy

- Insight: Visa and Diners Club customers receive higher average discounts.
- Action: Shift toward performance-based or discounts to protect margins.
- Impact: Balances competitiveness with profitability.

5. Quarterly Demand Planning

🔍 Insight: Q4 shows a 62.91% revenue surge and 4x order growth.

- Action: Prepare inventory and staffing for Q4 spikes. Use predictive analytics to forecast demand and avoid stockouts.
- Impact: Maximizes revenue potential and operational readiness.

6. Data-Driven Segmentation

- Next Step: Use clustering to segment customers by behavior, geography, and card type. Tailor offers and communication accordingly.
- Impact: Drives personalization, improves conversion rates, and supports strategic decision-making.